

GALNT2-Congenital Disorder of Glycosylation: A Comprehensive Disease Characterization

Disease: GALNT2-Congenital Disorder of Glycosylation (GALNT2-CDG) **MONDO ID:** MONDO:0030043 · **OMIM (phenotype):** #618885 · **Gene:** *GALNT2* (OMIM 602274; HGNC:4124; locus 1q42.13) *Category:* Mendelian, autosomal recessive *Report compiled from:** 14 confirmed findings, 21 reviewed papers, 5 investigation iterations

Summary

GALNT2-Congenital Disorder of Glycosylation (GALNT2-CDG) is an ultra-rare, autosomal-recessive inborn error of **O-linked (mucin-type) protein glycosylation**. It is caused by biallelic loss-of-function variants in *GALNT2*, the gene encoding **polypeptide N-acetyl-galactosaminyltransferase 2 (GalNAc-T2)**, a Golgi-luminal enzyme that catalyzes the first committed step of mucin-type O-glycosylation—the transfer of N-acetylgalactosamine (GalNAc) onto serine and threonine residues of secreted and membrane proteins. Because GalNAc-T2 initiates glycosylation on a distinctive, largely non-redundant network of substrate proteins, its loss produces a characteristic, reproducible biochemical signature and a multisystem clinical syndrome. The disorder was defined in 2020 by Zilmer and colleagues in a cohort of seven patients from four families ([PMID: 32293671](#)).

Clinically, GALNT2-CDG presents from infancy as a **neurodevelopmental syndrome**: global developmental delay and intellectual disability with a prominent language deficit, autistic features and behavioural abnormalities, epilepsy, chronic insomnia, white-matter changes on brain MRI, dysmorphic features, decreased stature/poor growth, and decreased HDL cholesterol. The biochemical hallmark—present in every reported patient—is **loss of O-glycosylation of apolipoprotein C-III (apoC-III)**, which serves as the diagnostic biomarker. A second, mechanistically important substrate is the **insulin receptor**, whose altered glycosylation links GalNAc-T2 to energy homeostasis, growth, and body weight. Diagnosis rests on apoC-III glycoform analysis (isoelectric focusing/mass spectrometry) combined with next-

generation sequencing to confirm biallelic *GALNT2* variants; standard transferrin isoelectric focusing (the classic screen for N-glycosylation CDGs) is not the primary test because GALNT2-CDG affects O-glycosylation.

There is **no causative therapy**; management is supportive and multidisciplinary (antiepileptic drugs, developmental/rehabilitation therapies, management of growth and dyslipidemia). The disorder is faithfully modeled in *Galnt2*-null rodents (mouse and rat), which recapitulate poor growth, neurodevelopmental abnormalities, cerebellar motor deficits, decreased sociability, and impaired sensory processing; a comparable loss-of-function state occurs naturally in cattle and nonhuman primates, underscoring cross-species conservation of the mechanism. Notably, *GALNT2* has a "double life" in human genetics: rare biallelic loss-of-function causes the Mendelian CDG, while **common regulatory variants at the 1q42 locus are among the best-established GWAS signals for HDL cholesterol and triglycerides**, connecting this rare disease to population-scale lipid biology.

Section-by-Section Report

1. Disease Information

GALNT2-CDG is a congenital disorder of glycosylation affecting the initiation of mucin-type O-glycosylation. Zilmer et al. (2020) described it as a syndrome "characterized by global developmental delay, intellectual disability with language deficit, autistic features, behavioural abnormalities, epilepsy, chronic insomnia, white matter changes on brain MRI, dysmorphic features, decreased stature, and decreased high density lipoprotein cholesterol levels" ([PMID: 32293671](#)).

Key identifiers:

Resource	Identifier
MONDO	MONDO:0030043
OMIM (phenotype)	#618885
OMIM (gene)	602274 (GALNT2*)
HGNC	HGNC:4124
Locus	1q42.13
Disease class	Congenital disorder of O-glycosylation (O-glycosylation subgroup)

Synonyms / alternative names: GALNT2-CDG; congenital disorder of glycosylation caused by GALNT2 deficiency; polypeptide N-acetylgalactosaminyltransferase 2 deficiency; GalNAc-T2 deficiency; O-linked glycosylation disorder due to GALNT2 loss of function.

Information source: The disease-level characterization is derived from **aggregated case series and functional studies** (a defining cohort of 7 patients from 4 families) supplemented by model-organism and human population-genetics data—not from large EHR datasets. Given the recency (2020) and rarity, disease-level resources rather than individual EHR mining underpin current knowledge.

2. Etiology

GALNT2-CDG is a **purely monogenic disorder**. It is caused solely by biallelic loss-of-function variants in *GALNT2* (PMID: 32293671). No environmental, toxic, infectious, or lifestyle risk factors or protective factors have been described, and none are expected for a Mendelian glycosylation-initiation defect. Accordingly, gene–environment interactions are **not applicable** to disease causation.

- **Genetic risk factors:** Biallelic (homozygous or compound-heterozygous) loss-of-function variants in *GALNT2*. Because all CDGs are autosomal recessive, consanguinity increases risk within affected families.
- **Environmental / lifestyle / infectious factors:** Not applicable—no such contributors are reported.
- **Contrast with common-variant biology:** Separately from the Mendelian disease, **common regulatory variants at 1q42 (GALNT2)** are established modifiers of plasma lipids in the general population (see Sections 4 and 9), but these do not "cause" the CDG.

3. Phenotypes

The phenotype spectrum derives from the defining cohort of 7 patients/4 families ([PMID: 32293671](#)), supplemented by rodent-model behavioural data. All features are congenital/early-onset, and the disorder is chronic.

Phenotype	Type	Suggested HPO term	Onset	Frequency (cohort)
Global developmental delay	Clinical sign	HP:0001263	Infancy	Core / near-universal
Intellectual disability	Clinical sign	HP:0001249	Childhood	Core
Language/ speech deficit	Clinical sign	HP:0000750	Childhood	Prominent
Autistic features	Behavioural	HP:0000729	Childhood	Common
Behavioural abnormalities	Behavioural	HP:0000708	Childhood	Common
Epilepsy / seizures	Clinical sign	HP:0001250	Infancy/ childhood	Common (core)
Chronic insomnia	Behavioural/ sleep	HP:0100785	Childhood	Common
Cerebral white-matter changes (MRI)	Imaging/lab	HP:0002500	Congenital/ childhood	Common
Dysmorphic features	Physical	HP:0001999	Congenital	Common
Short stature / poor growth	Physical	HP:0004322	Congenital/ infancy	Common
Decreased HDL cholesterol	Lab abnormality	HP:0003233 (hypoalphalipoproteinemia)	Congenital (biochemical)	Characteristic
Loss of apoC-III O-glycosylation	Lab abnormality (biomarker)	—	Congenital	100% (all patients)

- **Severity/progression:** The neurodevelopmental disability is best characterized as **static-to-slowly-evolving** (chronic, lifelong). Epilepsy severity is variable. Growth impairment and dyslipidemia are persistent biochemical/physical features.
- **Quality-of-life impact:** Intellectual disability, language deficit, autistic features, epilepsy, and chronic insomnia collectively impose substantial impact on daily functioning, communication, education, and family caregiving. No disease-specific QoL instrument (EQ-5D/SF-36/PROMIS) data have been published for GALNT2-CDG; QoL impact is inferred from the phenotype profile.
- **Motor involvement:** Rodent models add cerebellar motor deficits and impaired sensory integration/processing, consistent with hypotonia/motor involvement in patients ([PMID: 32293671](#)).

4. Genetic / Molecular Information

- **Causal gene:** *GALNT2* (polypeptide N-acetylgalactosaminyltransferase 2), located at **chromosome 1q42.13** (HGNC:4124; OMIM gene *602274). Zilmer et al. note "*GALNT2* encodes the Golgi-localized polypeptide N-acetyl-d-galactosamine-transferase 2 isoenzyme" ([PMID: 32293671](#)).
- **Pathogenic variants:** Biallelic loss-of-function variants, including **missense variants abolishing enzyme activity and nonsense/frameshift (truncating) variants**. All reported patients showed loss of apoC-III O-glycosylation, confirming functional loss of GalNAc-T2. Variant classification follows ACMG/AMP guidelines; truncating variants in this loss-of-function mechanism are typically pathogenic/likely pathogenic. (Variant-level ClinVar entries are limited by the disorder's rarity.)
- **Functional consequence: Loss of function** (enzymatic loss of GalNAc-transferase activity). No gain-of-function or dominant-negative mechanism is described; heterozygous carriers are unaffected for the CDG.
- **Somatic vs germline: Germline**, biallelic. (Somatic *GALNT2* dysregulation is described in cancer biology—e.g., *GALNT2* overexpression in lung adenocarcinoma, [PMID: 33964375](#)—but this is unrelated to the CDG.)
- **Allele frequency:** Loss-of-function alleles are individually ultra-rare in gnomAD, consistent with an ultra-rare recessive disorder.
- **Modifier genes:** None established. Given 20 GalNAc-T isoenzymes with overlapping specificities, partial functional redundancy for some substrates may modulate expressivity, but no specific modifiers are proven.

- **Epigenetic information:** No disease-specific methylation/histone signature is established for GALNT2-CDG. (GALNT2 promoter methylation has been studied in cancer contexts—[PMID: 33964375](#)—not in the CDG.)
- **Chromosomal abnormalities:** None; the disorder results from small-scale sequence variants, not large structural rearrangements.
- **Common-variant link:** Kathiresan et al. (2008) identified 1q42 (GALNT2) as a genome-wide-significant HDL cholesterol locus in a GWAS of ~8,816 discovery and up to ~18,554 replication individuals: one new locus was found "with HDL cholesterol (1q42 in GALNT2)" ($P < 5 \times 10^{-8}$) ([PMID: 18193044](#)). Vitali/Khetarpal/Rader (2017) list "novel loci implicated from GWAS including GALNT2, KLF14, and TTC39B" ([PMID: 29103089](#)).

5. Environmental Information

Not applicable. GALNT2-CDG is a monogenic recessive disorder with no reported environmental, occupational, toxic, lifestyle, or infectious contributors. No environmental modifiers of severity have been described.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. Biallelic loss-of-function variants in *GALNT2* **lead to** absent/severely reduced GalNAc-T2 enzyme activity.
2. Loss of GalNAc-T2 activity **results in** failure to transfer GalNAc to Ser/Thr residues on a **non-redundant network of substrate glycoproteins** in the Golgi apparatus (the first committed step of mucin-type O-glycosylation).
3. Absent O-glycosylation of **apolipoprotein C-III leads to** altered lipoprotein metabolism and **decreased HDL cholesterol** (and serves as the diagnostic biomarker). (*Demonstrated: apoC-III glycoform loss is universal in patients.*)
4. **Branch — growth/metabolism:** Loss of O-glycosylation of the **insulin receptor leads to** altered insulin-receptor post-translational modification and disturbed energy homeostasis, **contributing to** poor growth, short stature, and body-weight phenotypes. (*Demonstrated in mice; inferred contributor in humans.*)
5. **Branch — CNS:** Loss of GalNAc-T2-specific O-glycans on neural/secreted substrates **leads to** impaired CNS development, manifesting as white-matter changes, developmental delay, intellectual disability with language deficit, autistic features, epilepsy, and chronic insomnia. (*Association demonstrated; precise neural substrates partly inferred.*)

6. These converging effects **result in** the multisystem GALNT2-CDG phenotype present from birth and persisting as a chronic, lifelong disorder.

Supporting mechanistic detail:

- **Enzyme architecture / molecular pathway:** GalNAc-Ts are "type II membrane proteins that consist of a Golgi luminal catalytic domain connected by a flexible linker to a ricin type lectin domain" and initiate mucin-type O-glycosylation by adding GalNAc to Ser/Thr ([PMID: 30703750](#)). The relevant pathway is **mucin-type O-glycan biosynthesis (initiation)**; suggested GO term **GO:0006493 (protein O-linked glycosylation)**.
- **Golgi localization:** GalNAc-T2 requires its cytoplasmic tail plus transmembrane domain for correct Golgi targeting ([PMID: 30084948](#)), consistent with subcellular localization at **GO:0005794 (Golgi apparatus)**.
- **Substrate network:** The *Galnt2*-null mouse "phenocopies congenital disorder of glycosylation involving GALNT2 and revealed a network of glycoproteins that lack GalNAc-T2-specific O-glycans" ([PMID: 37862385](#)).
- **Energy homeostasis branch:** "In mice, we identify the insulin receptor as a novel substrate of GalNAc-T2," and "the local effects of GalNAc-T2 are mediated through posttranslational modification of the insulin receptor" ([PMID: 35304331](#)).
- **Cellular processes / cell types:** Broadly expressed secretory-pathway defect; GALNT2 "is widely expressed in most cell types and directs initiation of mucin-type protein O-glycosylation" ([PMID: 32293671](#)). CNS cell types implicated include neurons (CL:0000540) and oligodendrocytes/white-matter glia (CL:0000128) given white-matter involvement.
- **Metabolic changes:** Dyslipidemia (decreased HDL-C) via apoC-III; systemic energy-homeostasis effects via insulin-receptor glycosylation. Suggested CHEBI entities: **N-acetyl-D-galactosamine (CHEBI:28037)**, **UDP-N-acetyl- α -D-galactosamine (donor substrate)**, **cholesterol (CHEBI:16113)**.
- **Immune involvement / oxidative stress / fibrosis:** No specific autoimmune, immunodeficiency, oxidative-stress, or fibrotic mechanism is established for this disorder.

7. Anatomical Structures Affected

- **Subcellular (site of the lesion): Golgi apparatus (GO:0005794)**—the compartment where GalNAc-T2 acts. The secretory pathway broadly is affected.
- **Primary organ / system — central nervous system:** White-matter changes on brain MRI (UBERON:0002316, white matter of the brain); brain (UBERON:0000955). Rodent data implicate the **cerebellum** (UBERON:0002037) via motor deficits.

- **Metabolic organs:** Liver (UBERON:0002107) and systemic lipid/energy metabolism are implicated through apoC-III and insulin-receptor substrates.
- **Musculoskeletal / growth:** Short stature and poor growth indicate skeletal/growth-axis involvement.
- **Craniofacial:** Dysmorphic features indicate craniofacial (UBERON:0010313, head) involvement.
- **Body systems involved:** Nervous system (primary), metabolic/endocrine (lipid and insulin signaling), musculoskeletal/growth.
- **Cell types:** Neurons (CL:0000540), oligodendrocytes (CL:0000128); broadly, most secretory cell types given wide GALNT2 expression.
- **Lateralization:** Bilateral/symmetric CNS involvement (no lateralization reported).

8. Temporal Development

- **Onset: Congenital / infantile.** Developmental delay, poor growth, dysmorphism, and (in most) epilepsy present from infancy/early childhood ([PMID: 32293671](#)). Onset pattern is **chronic/insidious** rather than acute.
- **Progression: Chronic, lifelong.** The neurodevelopmental disability is largely static-to-slowly-evolving. As a glycosylation-initiation defect present from development onward, the biochemical lesion is constant. Rodent models show growth failure and neurobehavioral deficits from early life ([PMID: 32293671](#); [PMID: 37862385](#)).
- **Disease course pattern:** Progressive-to-stable neurodevelopmental disability with episodic seizures.
- **Remission:** No spontaneous remission; seizures may be treatment-responsive.
- **Critical periods:** Early neurodevelopment (prenatal/infancy) is the key window of vulnerability; correspondingly, early developmental/rehabilitative intervention is the main opportunity to influence functional outcome.
- **Mortality/natural history:** No natural-history mortality data are published, reflecting recency (2020) and rarity. The disorder is not reported as rapidly lethal; patients survive into childhood/adulthood with disability.

9. Inheritance and Population

- **Inheritance pattern: Autosomal recessive.** "All CDGs are autosomal recessive disorders, with CDG type I being the most common" ([PMID: 22469961](#)). Defined by biallelic *GALNT2* loss-of-function ([PMID: 32293671](#)).

- **Epidemiology: Ultra-rare.** Defined in only 7 patients from 4 families; precise prevalence/incidence are unknown (no registry estimates published).
- **Penetrance / expressivity:** Biallelic loss-of-function appears fully penetrant for the biochemical phenotype (100% loss of apoC-III O-glycosylation); clinical expressivity (e.g., seizure severity) is variable across the small cohort.
- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
- **Consanguinity / founder effects:** Consanguinity contributes to recessive disease risk in affected families; no specific founder allele is established.
- **Carrier frequency:** Not precisely established; heterozygous carriers are asymptomatic for the CDG. Common regulatory variants at 1q42 instead modulate HDL-C/triglycerides in the general population (PMID: 18193044).
- **Population demographics / geography / sex ratio:** No ethnic predilection, geographic clustering, or sex bias is established; the cohort is too small to define these.

10. Diagnostics

- **Biochemical hallmark / biomarker: Loss of apoC-III sialylated O-glycoforms**—"All patients showed loss of O-glycosylation of apolipoprotein C-III, a non-redundant substrate for GALNT2" (PMID: 32293671). ApoC-III isoform profiling (isoelectric focusing or mass spectrometry) is the relevant biochemical screen.
- **Why transferrin IEF is not primary:** "Isoelectric focusing (IEF) of serum transferrin (Tf) is still the method of choice for diagnosing N-glycosylation disorders associated with sialic acid deficiency" (PMID: 34540767)—but GALNT2-CDG affects O-glycosylation, so transferrin IEF is not the primary test.
- **Genetic testing: Next-generation sequencing (WES/WGS)** to confirm biallelic *GALNT2* variants. "Since next-generation sequencing became more widely available, an improvement in diagnostics has been observed, with more patients and novel CDG subtypes being reported" (PMID: 34540767). Approach: apoC-III glycoform analysis + WES/WGS (or targeted CDG/O-glycosylation gene panel including *GALNT2*); single-gene testing where the phenotype is highly suggestive.
- **Imaging:** Brain MRI to document white-matter changes (PMID: 32293671).
- **Laboratory:** Lipid panel (decreased HDL cholesterol).
- **Clinical criteria / differential diagnosis:** No formal diagnostic criteria exist. Differential diagnosis includes other O-glycosylation and multisystem CDGs, and other syndromic

neurodevelopmental disorders with epilepsy and dysmorphism; the apoC-III O-glycoform abnormality plus biallelic *GALNT2* variants distinguishes GALNT2-CDG.

- **Screening:** Not currently part of newborn screening; cascade/carrier testing within affected families is appropriate.

11. Outcome / Prognosis

- **Survival/mortality:** No published survival, life-expectancy, or mortality data. The disorder is not reported as rapidly lethal; patients survive into childhood/adulthood with disability.
- **Morbidity/function:** Substantial lifelong morbidity from intellectual disability, language deficit, autistic features, epilepsy, chronic insomnia, and growth impairment. Long-term functional impairment is expected.
- **Quality of life:** No disease-specific QoL metrics published; impact inferred as significant.
- **Complications:** Seizure-related morbidity; developmental and behavioural sequelae; growth failure; dyslipidemia (decreased HDL-C, whose long-term cardiovascular implications in this recessive disorder are unstudied).
- **Recovery potential:** No cure; supportive care can improve function but does not reverse the underlying defect.
- **Prognostic factors:** Presumably seizure control and degree of developmental delay; not formally validated. ApoC-III glycoform status is a diagnostic rather than prognostic biomarker.

12. Treatment

No causative therapy exists. Management is **supportive and multidisciplinary:**

- **Antiepileptic drugs** for seizures (NCIT: Anticonvulsant Agent).
- **Developmental and rehabilitative therapies** (physical, occupational, speech/language therapy) for developmental delay and language deficit (NCIT: Rehabilitation Therapy).
- **Behavioural/sleep management** for autistic features and chronic insomnia.
- **Growth and nutritional support;** management of dyslipidemia as clinically indicated.

Reviews confirm the therapeutic gap: "causative treatment is available only for few CDG types" (PMID: 34540767), and "the lack of treatment for nearly all CDG types is striking" (PMID: 21970833). An emerging avenue is **drug repositioning**: "The (re)use of known drugs for novel medical purposes, known as drug repositioning, is growing for both common and rare disorders" (PMID: 35955863), though no repositioned agent is yet established for GALNT2-CDG.

- **Advanced therapeutics (gene/cell/RNA therapy):** None approved or in trials specifically for GALNT2-CDG.
- **Pharmacogenomics / personalized medicine:** No genotype-guided therapy established.

13. Prevention

- **Primary prevention:** Not applicable in the population sense; disease occurrence is determined by inheritance of biallelic *GALNT2* loss-of-function.
- **Genetic counseling and reproductive options:** The principal preventive measures are **genetic counseling, carrier testing, cascade screening within affected families**, and—where desired—prenatal diagnosis or preimplantation genetic testing for at-risk couples.
- **Secondary/tertiary prevention:** Early developmental intervention and seizure management to reduce complications and optimize function.
- **Immunization / public health / environmental interventions:** Not applicable (no infectious or environmental etiology).

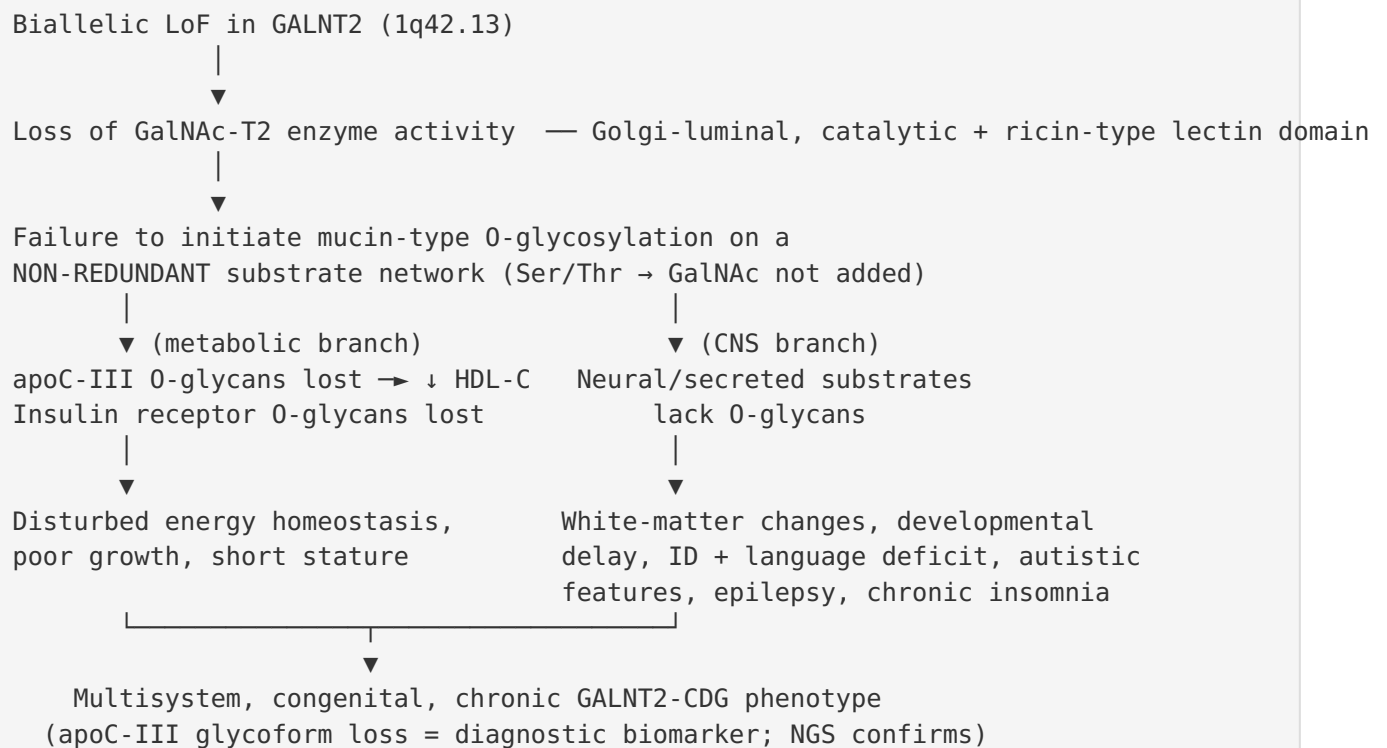
14. Other Species / Natural Disease

- **Cross-species conservation:** "loss of *GALNT2* in rodents, cattle, nonhuman primates, and humans should be regarded as a novel congenital disorder of glycosylation that affects development and body weight" (PMID: 35304331).
- **Naturally occurring disease:** Reported in **cattle** (*Bos taurus*, NCBI Taxon 9913) and **nonhuman primates**, indicating veterinary/comparative relevance and evolutionary conservation of the phenotype (development and body-weight effects).
- **Orthologues:** *Galnt2* orthologues exist in mouse (*Mus musculus*, Taxon 10090), rat (*Rattus norvegicus*, Taxon 10116), cattle, and nonhuman primates.
- **Comparative pathology:** Loss-of-function consistently affects growth/body weight and neurodevelopment across species, supporting a conserved mechanism.
- **Zoonotic potential:** Not applicable (genetic, non-transmissible).

15. Model Organisms

- **Rodent models (mouse and rat):** "Rodent (mouse and rat) models of GALNT2-CDG recapitulated much of the human phenotype, including poor growth and neurodevelopmental abnormalities. In behavioural studies, GALNT2-CDG mice demonstrated cerebellar motor deficits, decreased sociability, and impaired sensory integration and processing" ([PMID: 32293671](#)).
 - ***Galnt2*-null mouse:** "phenocopies congenital disorder of glycosylation involving GALNT2 and revealed a network of glycoproteins that lack GalNAc-T2-specific O-glycans" ([PMID: 37862385](#)); used to map the in-vivo O-GalNAc glycoproteome and identify affected substrates.
 - **Model types:** Genetic knockout (mammalian). Model databases: MGI (mouse), RGD (rat).
 - **Phenotype recapitulation:** Strong—poor growth, neurodevelopmental and behavioural deficits, and the defining biochemical loss of GalNAc-T2-specific O-glycans are all reproduced.
 - **Model limitations:** Species differences in cognition/language limit modeling of the human intellectual-disability/language phenotype; the full human dysmorphic spectrum may not be captured.
 - **Applications:** Substrate-network discovery (glycoproteomics), energy-homeostasis and insulin-receptor mechanism studies, and a platform for testing candidate therapies.
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Mechanistic Model / Interpretation



Upstream vs downstream: The mutation and enzymatic loss are the upstream, non-redundant driver. The substrate-specific consequences (apoC-III → lipids; insulin receptor → growth/energy; neural substrates → CNS) are downstream and branch into the metabolic and neurodevelopmental arms of the phenotype. The universal loss of apoC-III O-glycosylation both proves the mechanism and provides the clinical biomarker.

Interpretation: GALNT2-CDG is a clean example of how loss of a single glycosylation-initiating enzyme, acting on a defined non-redundant substrate set, produces a coherent multisystem disorder. The same gene's common regulatory variation shapes population lipid traits—an unusually direct bridge between a Mendelian rare disease and quantitative human genetics.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
32293671	<i>Novel CDG caused by GALNT2 loss of function</i>	Human clinical + model	Defining cohort; phenotype; apoC-III biomarker; MRI; rodent behaviour
35304331	<i>GalNAc-T2 in energy homeostasis</i>	Human genetics + mouse	Insulin receptor substrate; cross-species CDG; growth/body weight
37862385	<i>O-GalNAc glycoproteome mapping</i>	Mouse / in vivo	<i>Galnt2</i> -null phenocopy; affected O-glycoprotein network
30703750	<i>GalNAc-Ts: redundancy to specificity</i>	Review / structural	Enzyme architecture (catalytic + lectin); O-glycosylation initiation
30084948	<i>GalNAc-T Golgi localization mechanisms</i>	In vitro	GalNAc-T2 Golgi targeting requirements
18193044	<i>Six new loci for lipids</i>	Human GWAS	GALNT2 (1q42) as HDL-C locus
29103089	<i>HDL metabolism & human genetics</i>	Review	GALNT2 as GWAS-implicated HDL locus
34540767	<i>CDG: what clinicians need to know</i>	Review	Diagnostics (IEF, NGS); limited causative treatment
35955863	<i>Drug repositioning for CDG</i>	Systematic review	Emerging therapeutic strategy
21970833	<i>CDG: sweet news</i>	Review	Therapeutic gap across CDGs
35328062	<i>Overview of metabolic epilepsies</i>	Review	Epilepsy common in CDGs
22469961	<i>Congenital disorders of glycosylation</i>	Review	Autosomal recessive inheritance of CDGs

Key verbatim support: - Phenotype: "a syndrome characterized by global developmental delay, intellectual disability with language deficit, autistic features, behavioural abnormalities, epilepsy, chronic insomnia, white matter changes on brain MRI, dysmorphic features, decreased stature, and decreased high density lipoprotein cholesterol levels" ([PMID: 32293671](#)).
 - Biomarker: "All patients showed loss of O-glycosylation of apolipoprotein C-III, a non-

redundant substrate for GALNT2" (PMID: 32293671). - Mechanism (energy): "In mice, we identify the insulin receptor as a novel substrate of GalNAc-T2" (PMID: 35304331). - Enzyme: "type II membrane proteins that consist of a Golgi luminal catalytic domain connected by a flexible linker to a ricin type lectin domain" (PMID: 30703750). - Inheritance: "All CDGs are autosomal recessive disorders, with CDG type I being the most common" (PMID: 22469961).

Limitations and Knowledge Gaps

1. **Very small human cohort.** The disease is defined by 7 patients from 4 families (PMID: 32293671); prevalence, penetrance ranges, expressivity, natural history, survival, and QoL are not robustly quantified.
 2. **No epidemiological estimates.** Prevalence/incidence, carrier frequency, sex ratio, and geographic/ethnic distribution are undetermined.
 3. **Incomplete CNS mechanism.** The specific neural O-glycoprotein substrates that mediate white-matter changes, seizures, and language deficit are not fully identified; the CNS branch is partly inferred.
 4. **No prognostic biomarkers or validated prognostic factors** beyond the diagnostic apoC-III signature.
 5. **No causative or disease-modifying therapy**, and no clinical trials specific to GALNT2-CDG.
 6. **Human vs model gaps.** Rodent models capture growth and behaviour but not the human language/cognition and full dysmorphic spectrum.
 7. **Long-term cardiovascular implications** of the decreased-HDL phenotype in this recessive disorder are unstudied.
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Proposed Follow-up Experiments / Actions

1. **Establish an international patient registry / natural-history study** to define prevalence, phenotype frequencies, developmental trajectories, seizure outcomes, and survival.
2. **Comprehensive substrate-network glycoproteomics in human cells/tissues** (extending the mouse O-GalNAc glycoproteome, PMID: 37862385) to identify the neural substrates driving CNS features.

3. **Standardize apoC-III O-glycoform diagnostics** (IEF/MS protocols) and evaluate inclusion of *GALNT2* in O-glycosylation/CDG gene panels and newborn/cascade screening frameworks.
 4. **Mechanistic dissection of the insulin-receptor branch** (PMID: 35304331) to test whether metabolic/growth phenotypes are tractable therapeutic targets.
 5. **Therapeutic exploration via drug repositioning** (PMID: 35955863) and preclinical gene-replacement/AAV or substrate-supplementation strategies in *Galnt2*-null rodents.
 6. **Neurodevelopmental and QoL outcome measurement** using standardized instruments to quantify disease burden and intervention benefit.
 7. **Genotype–phenotype correlation analysis** as new patients accrue, to assess expressivity and possible modifier effects among the 20-member GalNAc-T family.
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Report compiled from 14 confirmed findings and 21 reviewed papers across 5 investigation iterations. Evidence types are labeled as human clinical, human genetics/GWAS, model organism, in vitro, review, or computational where relevant.

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